

????????????????

1. ??

??

2. ?????????

Unity?????????Y????????? d ?????????????????????????????Y??? y ????????? d = -y ??????????y = 0 ???y < 0 ?????
????Unity??? y = -10 ????? d = 10 m ?????

3. ?????????

????????CSV????????CSV????????????depth_m ?????[m]?temperature_c ?????????[degC]???

```
depth_m,temperature_c
0,20
5,18
10,15
20,10
```

???????? T(d) ??CSV????????-??

d_i ? $d_{(i+1)}$????? d ????? $T(d) = T_i + ((d - d_i) / (d_{(i+1)} - d_i)) * (T_{(i+1)} - T_i)$???

4. ?????

????????????????????????????????????

4.1 ?????

???????????????????? FreshWater ????????????? c ??? T ??? d ?????????

$c(T,d) = c_0(T) + 0.0163 d$

$c_0(T) = 1402.388 + 5.03830T - 5.81090e-2 T^2$
 $+ 3.3432e-4 T^3 - 1.47797e-6 T^4$
 $+ 3.1419e-9 T^5$

??? T ???[degC]?d ???[m]?c ???[m/s]???0.0163 d ?????????????

4.2 ?????

???????????? Mackenzie1981 ????????????? T??? S??? d ?????

$c(T,S,d) = 1448.96 + 4.591T - 5.304e-2 T^2$
 $+ 2.374e-4 T^3 + 1.340(S-35)$
 $+ 1.630e-2 d + 1.675e-7 d^2$
 $- 1.025e-2 T(S-35) - 7.139e-13 T d^3$

S ???[PSU]???????? S = 0???????? S ???35????

5. ???????

????????????????3????????

None: ???

RangeOnly: ?????????????????????????????????????

RefractionAndRange: ?????????????????????????

6. ?????????

???????????? ?s ????????????????????? c_i???????? ?s_j ????????????? t ? t = sum(?s_i / c_i) ???????

???????????????????? R_geo = sum(?s_i) ????????????????????? c_ref ????????? R_acoustic = c_ref * t ???
???

??CSV????????????range_m?geometric_range_m?acoustic_range_m?travel_time_s ??????

7. ???????

???????????????????????????????????? c = c(d) ?????????????????

????????????Snell??

???????????????? ? cos(?) / c(d) = constant ????????????????? c1???????? c2???????? cos(?1) ???
???????? cos(?2) = (c2 / c1) cos(?1) ??????

cos(?2) >= 1 ?????????????????????????????????

8. ?????

??

??

??

9. ???????

????????????????????????????None?RangeOnly?RefractionAndRange ?3????????????????